

IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 05 REPORT

Enterprise Services Infrastructure Deployment

Centralized DHCP · DNS · HTTP/HTTPS · FTP/TFTP · NTP · Syslog · SNMP

Field	Detail
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Assigned To	Network Administration Interns
Tools Used	Cisco Packet Tracer
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1. Objectives

- Design and deploy a centralized enterprise services infrastructure supporting multiple departments.
- Configure and integrate DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP on one authoritative server.
- Use DHCP relay (ip helper-address) so every VLAN reaches that one central server instead of running per-router DHCP pools.
- Verify every service actually works end-to-end, not just that it's toggled on.
- Document a real troubleshooting scenario from getting these services to interoperate.

2. Network Design

This reuses the same three-department layout as Week 4 (HR, Finance, IT) — same company, ongoing internship — but this week's point is different: instead of securing traffic between departments, the goal is centralizing every shared service onto one authoritative server so there's a single source of truth for addressing, naming, time, and logging across the whole network.

The key design decision: DHCP is NOT configured locally on R1 this time. Every department VLAN carries an 'ip helper-address' pointing at the central server, which relays DHCP broadcasts across the Layer 3 boundary to the one real DHCP service. This is how DHCP actually gets centralized in a real enterprise — a router will not forward a DHCP broadcast on its own without this configured, since broadcasts don't cross subnets by default.

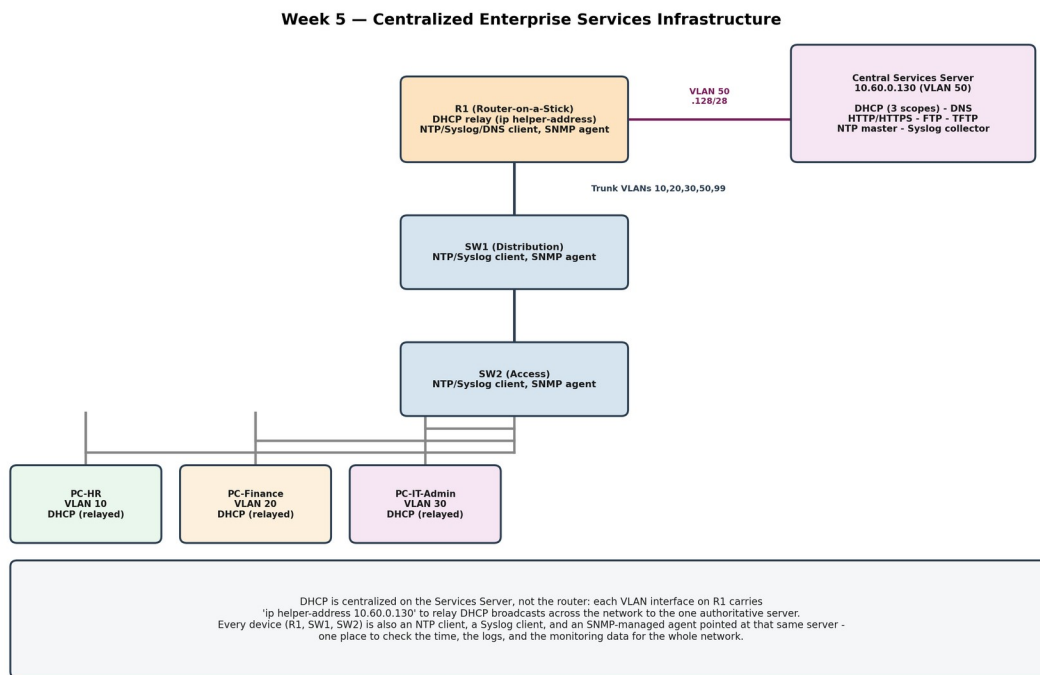


Figure 1 — Topology: three department VLANs, a dedicated Server Farm VLAN, and the centralized services server.

3. IP Addressing Table (VLSM)

Base network: 10.60.0.0/16, with a dedicated Server Farm VLAN this time in addition to the Management VLAN.

VLAN	Segment	Subnet (CIDR)	Usable Range	Gateway
10	HR	10.60.0.0/26	10.60.0.1 - .62	10.60.0.1 (R1 sub-if)
20	Finance	10.60.0.64/27	10.60.0.65 - .94	10.60.0.65 (R1 sub-if)
30	IT	10.60.0.96/27	10.60.0.97 - .126	10.60.0.97 (R1 sub-if)
50	Server Farm	10.60.0.128/28	10.60.0.129 - .142	10.60.0.129 (R1 sub-if)
99	Management	10.60.0.144/29	10.60.0.145 - .150	10.60.0.145 (R1 sub-if)

Central Services Server: 10.60.0.130/28 (static). SW1 management: 10.60.0.146/29. SW2 management: 10.60.0.147/29.

4. Central Server — Service Configuration

Packet Tracer's Server-PT device is configured through its Services tab (GUI forms), not the IOS CLI — worth knowing before you go looking for config commands that don't exist for this part. The table below is what to set in each service tab.

Service	Setting
DHCP	3 pools: HR_POOL (10.60.0.0/26, start .6, gateway .1, DNS .130), FIN_POOL (10.60.0.64/27, start .70, gateway .65, DNS .130), IT_POOL (10.60.0.96/27, start .102, gateway .97, DNS .130). Service: On.
DNS	A records: www.itsimplera.local -> 10.60.0.130, ftp.itsimplera.local -> 10.60.0.130, services.itsimplera.local -> 10.60.0.130. Service: On.
HTTP/HTTPS	Edit index.html to a simple company homepage; enable HTTPS (Packet Tracer auto-generates the certificate). Service: On for both.
FTP	Users: admin / Admin@12345 (full RWDNL permissions), backup / Backup@12345 (write-only, for config backups). Service: On.
TFTP	No authentication by design — add a placeholder file so a get/put test has something to transfer. Service: On.
NTP	Set as authoritative time source (enable NTP service, set the server's own clock correctly first — everything else syncs from this).
Syslog	Service: On. This is the collector every device's 'logging host' command points to.

SNMP note: Packet Tracer's Server-PT does not include a full SNMP manager/MIB browser — there's no polling dashboard to screenshot. 'SNMP verification' here means configuring the community strings and trap destination on each network device (Section 5) and confirming with 'show snmp' that the agent is active and pointed at the server, which is what's actually achievable in this tool.

5. Router & Switch Configurations

R1

```
hostname R1
!
service password-encryption
!
ip domain-name itsimplera.local
ip name-server 10.60.0.130
!
interface GigabitEthernet0/0
 no shutdown
!
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 10.60.0.1 255.255.255.192
 ip helper-address 10.60.0.130
!
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 10.60.0.65 255.255.255.224
 ip helper-address 10.60.0.130
!
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 10.60.0.97 255.255.255.224
 ip helper-address 10.60.0.130
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 10.60.0.129 255.255.255.240
!
interface GigabitEthernet0/0.99
 encapsulation dot1Q 99
 ip address 10.60.0.145 255.255.255.248
!
ntp server 10.60.0.130
!
logging host 10.60.0.130
logging trap informational
logging source-interface GigabitEthernet0/0.99
!
snmp-server community ITSIMPLERA_R0 RO
snmp-server community ITSIMPLERA_RW RW
snmp-server host 10.60.0.130 version 2c ITSIMPLERA_R0
snmp-server enable traps
snmp-server location HQ-DataCloset
snmp-server contact netadmin@itsimplera.local
!
end
```

SW1 (Distribution)

```
hostname SW1
!
service password-encryption
!
ip domain-name itsimplera.local
ip name-server 10.60.0.130
!
vlan 10
 name HR
```

```

vlan 20
  name Finance
vlan 30
  name IT
vlan 50
  name ServerFarm
vlan 99
  name Management
!
interface Vlan99
  ip address 10.60.0.146 255.255.255.248
  no shutdown
!
ip default-gateway 10.60.0.145
!
interface GigabitEthernet0/1
  description Trunk to R1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,50,99
!
interface GigabitEthernet0/2
  description Trunk to SW2
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,99
!
ntp server 10.60.0.130
!
logging host 10.60.0.130
logging trap informational
!
snmp-server community ITSIMPLERA_R0 R0
snmp-server host 10.60.0.130 version 2c ITSIMPLERA_R0
!
end

```

SW2 (Access)

```

hostname SW2
!
service password-encryption
!
ip domain-name itsimplera.local
ip name-server 10.60.0.130
!
vlan 10
  name HR
vlan 20
  name Finance
vlan 30
  name IT
vlan 99
  name Management
!
interface Vlan99
  ip address 10.60.0.147 255.255.255.248
  no shutdown
!
ip default-gateway 10.60.0.145
!
interface GigabitEthernet0/1
  description Trunk to SW1
  switchport mode trunk
  switchport trunk allowed vlan 10,20,30,99
!
interface FastEthernet0/1

```

```
description PC-HR
switchport mode access
switchport access vlan 10
!
interface FastEthernet0/2
description PC-Finance
switchport mode access
switchport access vlan 20
!
interface FastEthernet0/3
description PC-IT-Admin
switchport mode access
switchport access vlan 30
!
ntp server 10.60.0.130
!
logging host 10.60.0.130
logging trap informational
!
snmp-server community ITSIMPLERA_R0 R0
snmp-server host 10.60.0.130 version 2c ITSIMPLERA_R0
!
end
```

6. FTP / TFTP Configuration Backup & Restore

This is the practical reason FTP and TFTP matter beyond file transfer demos — using the central server to back up and restore device configurations:

```
! FTP config backup (authenticated)
R1# copy running-config ftp://admin:Admin%4012345@10.60.0.130/R1-config
!
! TFTP config backup (unauthenticated)
R1# copy running-config tftp://10.60.0.130/R1-config
!
! TFTP config restore
R1# copy tftp://10.60.0.130/R1-config running-config
```

[Insert screenshots of a successful FTP backup (with login prompt) and a successful TFTP backup/restore]

7. Verification — the 10 Required Screenshots

The assignment asks for 10 screenshots; these map directly to that list, one per line item below.

```
R1# show ip dhcp binding          (or check DHCP scope/leases on the server GUI)
PC-HR> ipconfig /all             (confirm address, gateway, DNS came from the server's
scope)
PC-HR> nslookup www.itsimplera.local
PC-HR> ping www.itsimplera.local
PC-HR> (browser) http://www.itsimplera.local and https://www.itsimplera.local
PC-IT-Admin> (browser or CLI ftp client) connect to 10.60.0.130, login, put/get a file
R1# copy running-config tftp://10.60.0.130/R1-config
R1# show ntp status
R1# show ntp associations
R1# show logging | include Syslog
R1# show snmp
SW1# show snmp
PC-HR> ping 10.60.0.130
PC-HR> tracert 10.60.0.130
```

#	Service	What to Screenshot
1	DHCP	A PC's 'ipconfig /all' showing an address from the correct department pool, or the server's DHCP lease table
2	DNS	'nslookup www.itsimplera.local' or a successful 'ping www.itsimplera.local' resolving to 10.60.0.130
3	HTTP	Browser successfully loading http://www.itsimplera.local
4	HTTPS	Browser successfully loading https://www.itsimplera.local (padlock/certificate visible)
5	FTP	Successful login and file transfer to/from the FTP service
6	TFTP	Successful 'copy tftp://...' backup or restore
7	NTP	'show ntp status' / 'show ntp associations' showing synchronized state
8	Syslog	A log event appearing on the server's syslog collector after triggering one (e.g. an interface flap)
9	SNMP	'show snmp' on a device confirming the agent is active with the configured community/host
10	Connectivity	Ping and traceroute from a department PC to the server, confirming reachability across the relay path

8. Troubleshooting Report

Worked example of the kind of entry this section needs — replace with your own real issue once you've built and tested this:

Symptom	Diagnosis Steps	Root Cause	Resolution
PC-Finance never receives a DHCP lease, stays on APIPA	Checked cabling and VLAN assignment (correct); checked the server's DHCP service was On (it was); checked R1's sub-interface config for Gi0/0.20	'ip helper-address' was missing on the Finance sub-interface — it had been added to HR and IT but not Finance, so Finance's DHCP broadcasts had nowhere to be relayed to	Added 'ip helper-address 10.60.0.130' to Gi0/0.20; PC-Finance got a lease on the next DHCP request
[Your real scenario]	[What you checked, in order]	[What was actually wrong]	[What you changed to fix it]

9. Challenges I Ran Into

[Replace with what actually gave you trouble. Worth checking: did DHCP relay work on the first try, or did a VLAN's helper-address get missed like the example above? Did HTTPS need a certificate step you didn't expect? Did the FTP write permissions need adjusting before a backup would actually save? Did NTP sync take longer than expected, or need the server's own clock set first?]

10. Key Learning Outcomes

- I understand why DHCP broadcasts don't cross subnets on their own, and what 'ip helper-address' actually does about that.
- I can explain the practical difference between FTP (authenticated, reliable) and TFTP (open, lightweight) and why both still exist in real networks.
- I configured NTP, Syslog, and SNMP as a coherent set — not separate features, but the three things that make a network's state (time, events, health) checkable from one place.
- I know what Packet Tracer's Server-PT can and can't actually demonstrate for SNMP, instead of assuming a feature exists because the protocol does.
- I have a real example of a per-VLAN configuration detail (one missing helper-address) breaking service for exactly one department while the others worked fine — a useful pattern to recognize faster next time.

11. Conclusion

Centralizing services is a tradeoff, not a pure win: it's more efficient to manage one DHCP scope set, one DNS zone, and one log collector than to duplicate that configuration on every router — but it also means every department now depends on that one server and the relay path reaching it. This week was really about building that dependency correctly (helper-address on every VLAN, every device pointed at the same NTP/Syslog/SNMP target) and then proving each piece works instead of assuming a GUI toggle labeled 'On' means the service is actually reachable.

Appendix A — Submission Checklist

Deliverable	Status
Cisco Packet Tracer (.pkt) project file	[Build using Sections 4-5, then attach]
Complete topology with devices, servers, services labeled	Included — Section 2, Figure 1
Configuration files/screenshots for all routers, switches, servers	Complete — Sections 4-5
DHCP verification	[Insert in Section 7, item 1]
DNS verification	[Insert in Section 7, item 2]
HTTP/HTTPS verification	[Insert in Section 7, items 3-4]
FTP and TFTP verification (backup/restore)	[Insert in Section 6]
NTP verification	[Insert in Section 7, item 7]
Syslog verification	[Insert in Section 7, item 8]
SNMP verification	[Insert in Section 7, item 9]
Connectivity test results	[Insert in Section 7, item 10]
Troubleshooting report	Template in Section 8 — replace with your real scenario
Final documentation (10+ pages)	This document
LinkedIn Post	[See draft provided separately]
GitHub Repository	[See README provided separately]